# Instructions · How to Solve a Medical Detective Case

**Medical Detective Workbook · ages 11–14 · The C-E-R Framework**

Every lesson in your workbook is a real medical mystery — cholera in London, germs in space, a boy bitten by a rabid dog. Your job is to solve each one, but a detective can't just guess. You have to **prove** your answer, the way a real scientist does, using three tools: **Claim, Evidence, Reasoning** — C-E-R for short.

This sheet shows you exactly how to fill in every CER table, step by step. Lesson 1 — the Soho cholera case — is used as a worked example, so you can see what a finished answer actually looks like.

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## The C-E-R Framework

**CLAIM** — A clear, direct sentence that answers the investigation question. No "maybe," no "it could be."

**EVIDENCE** — Specific facts, numbers, or observations taken from the case file. Not your own opinion, and nothing you haven't actually read.

**REASONING** — The science that explains *why* your evidence proves your claim, and not some other claim.

Think of it like a court case. Your **claim** is the verdict. Your **evidence** is what was actually found at the scene. Your **reasoning** is why that evidence points to that verdict — and no other one.

## Step 1 · Read before you write

1. **Read the Case File.** This is the mystery. Don't skip it to get to the table faster — the answer is hidden in there.
2. **Read every line of the Case Clues & Evidence Checklist.** These clues are the only evidence you're allowed to use. Don't add facts you remember from somewhere else.
3. **Read the Investigation Prompt twice.** This is the exact question your claim has to answer — nothing wider, nothing narrower.

## Step 2 · Fill in your CER table, in order

1. **Write your CLAIM first.** One sentence. A direct answer to the Investigation Prompt.
2. **List your EVIDENCE.** Pick out the specific clues from the checklist that back up your claim. Point to numbers, places, and people — not vague ideas.
3. **Explain your REASONING.** Say *why* that evidence proves the claim, using the science mechanism from the case — don't just repeat the evidence in different words.
4. **Use the CLIL sentence frame** under the table to check your English. Fill in the brackets with your own claim, evidence, and reasoning.

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## Worked example · Lesson 1, Soho 1854

**Investigation Prompt**

What was the true vector of transmission for the 1854 Soho cholera outbreak?

**CLAIM**

Cholera in Soho spread through water from the *Broad Street pump*, not through bad air.

**EVIDENCE**

Deaths clustered around the *Broad Street pump*. The *brewery workers* next door drank beer instead of pump water and stayed healthy. The workhouse had its own private well and was barely affected. A widow in *Hampstead*, miles away, died after having pump water delivered to her home.

**REASONING**

People got sick based on which water they drank, not on how close they lived to the sick. The brewery workers lived right next to the pump but didn't drink from it, and stayed well; the Hampstead widow lived far away but drank from it, and died. If the disease travelled through the air, distance would be what mattered — instead, the water source is what mattered, every time.

**Sentence frame, filled in**

*"I claim that cholera spreads via **water** because the data shows **deaths clustered around the Broad Street pump while nearby beer-drinkers stayed healthy**, which means that **the pump water itself was contaminated, not the air around it**."*

### Three rules that matter

- **Your CLAIM is one sentence, not a paragraph.** If it needs the word "and" to connect two ideas, you probably have two claims — pick the one the prompt actually asks for.
- **Your EVIDENCE must come from the Case Clues checklist.** Don't add outside facts you remember, looked up, or guessed — a detective only uses what's in the file.
- **Your REASONING answers "why," not "what."** If your reasoning just repeats the evidence in different words, go one step deeper: what's the actual mechanism connecting them?

## What to hand in, for each lesson

1. **Your completed CER table** — Claim, Evidence, and Reasoning, all three filled in.
2. **The CLIL sentence frame,** filled in with your own words, not just the example from this sheet.
3. **Any Local Context answer,** if that lesson's case file asks one.

## Checking your own work before you hand it in

- Put your **Investigation Prompt** and your **CLAIM** side by side. Does the claim actually answer that exact question?
- Does your **EVIDENCE** name specific clues — a number, a place, a person? "There was evidence that..." is not evidence.
- Does your **REASONING** explain the mechanism in your own words, rather than copying the evidence sentence again?
- Did you fill in the **sentence frame**, and does it read as a real, sensible sentence out loud?